

$$y^2 =$$

1. Continuity

Definition

A function f is continuous at a number a if

$$\lim_{x \rightarrow a} f(x) = f(a)$$

This definition actually implicitly requires three things:

- ① $f(a)$ is defined
- ② $\lim_{x \rightarrow a} f(x)$ exists
- ③ $\lim_{x \rightarrow a} f(x) = f(a)$

A function f is continuous from the right at a number a if $\lim_{x \rightarrow a^+} f(x) = f(a)$

A function f is continuous from the left at a number a if $\lim_{x \rightarrow a^-} f(x) = f(a)$

A function is continuous on an interval if it is continuous at **every number in the interval**. (If f is defined only on one side of an endpoint of the interval, we understand *continuous* at the endpoint to mean *continuous from the right* or *continuous from the left*.)

If f and g are continuous at a and c is a constant, then the following functions are also continuous at a :

① $f + g$

② $f - g$

③ cf

④ fg

⑤ $\frac{f}{g}$ (if $g(a) \neq 0$)

The following types of functions are continuous at every number in their domain(s):

- ① polynomials
- ② rational functions
- ③ root functions
- ④ (inverse) trigonometric functions
- ⑤ exponential functions
- ⑥ logarithmic functions

If f is continuous at b and $\lim_{x \rightarrow a} g(x) = b$, then $\lim_{x \rightarrow a} f(g(x)) = f(b)$.

In other words,

$$\lim_{x \rightarrow a} f(g(x)) = f(\lim_{x \rightarrow a} g(x))$$

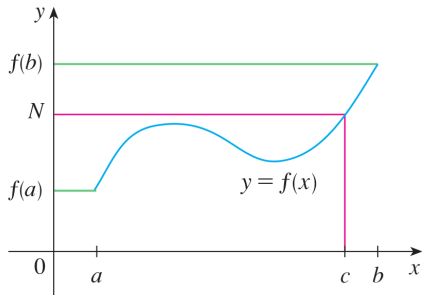
If g is continuous at a and f is continuous at $g(a)$, then $f(g(x))$ is continuous at a .

"A continuous function of a continuous function is a continuous function."

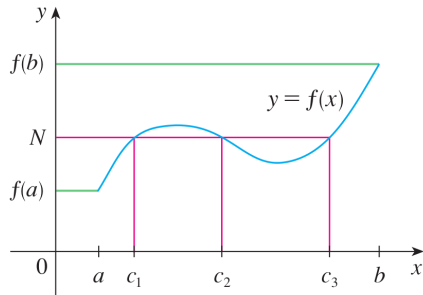
Theorem 5-The Intermediate Value Theorem

Suppose that f is **continuous on the closed interval $[a, b]$** and let N be any number between $f(a)$ and $f(b)$, where $f(a) \neq f(b)$. Then there exists a number c in (a, b) such that $f(c) = N$.

Note that the value N can be taken on once or more than once.

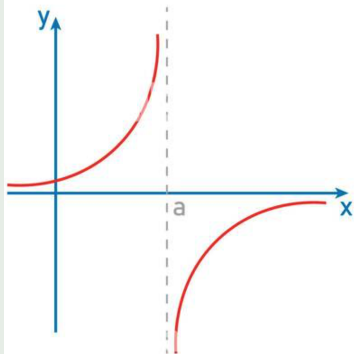


(a)

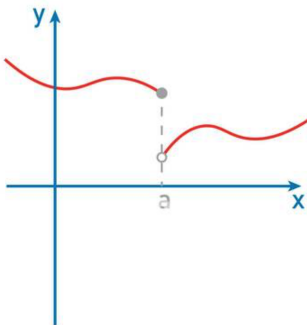


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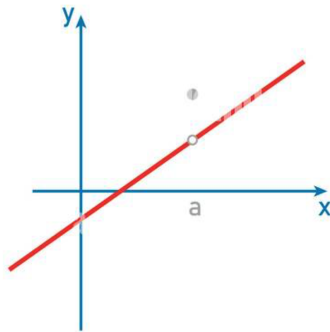
- ① removable discontinuity
- ② infinite discontinuity
- ③ jump discontinuity



Infinite Discontinuity



Jump Discontinuity



Removable Discontinuity

Use the Intermediate Value Theorem to show that there is a root of the given equation in the specified interval.

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Solution:

$\sqrt[3]{x} = 1 - x$ has a root on $(0, 1)$ equals to $f(x) = \sqrt[3]{x} + x - 1 = 0$ has a solution on $(0, 1)$. Since $f(0) = -1$, $f(1) = 1$, and 0 is between -1 and 1, there must be a point c such that $f(c) = 0$.

Let $f(x) = \frac{e^{\frac{1}{x}} - 1}{e^{\frac{1}{x}} + 1}$. What kind of discontinuity is $x = 0$?

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Solution:

$$\lim_{x \rightarrow 0^-} f(x) = \frac{0 - 1}{0 + 1} = -1$$

$$\lim_{x \rightarrow 0^+} f(x) = \frac{1 - \frac{1}{e^{\frac{1}{x}}}}{1 + \frac{1}{e^{\frac{1}{x}}}} = 1$$

Jump discontinuity.

Find the values of a and b that make f continuous everywhere.

$$f(x) = \begin{cases} \frac{x^2 - 4}{x - 2} & x < 2 \\ ax^2 - bx + 3 & 2 \leq x < 3 \\ 2x - a - b & x \geq 3 \end{cases} \quad (1)$$

Find the values of a and b that make f continuous everywhere.

$$f(x) = \begin{cases} \frac{x^2 - 4}{x - 2} & x < 2 \\ ax^2 - bx + 3 & 2 \leq x < 3 \\ 2x - a - b & x \geq 3 \end{cases} \quad (2)$$

Solution:

$$\lim_{x \rightarrow 2^-} f(x) = \frac{(x-2)(x+2)}{(x-2)} = 4$$

$$f(2) = 4a - 2b + 3$$

$$\lim_{x \rightarrow 3^-} f(x) = 9a - 3b + 3$$

$$f(3) = 6 - a - b$$

$$\text{Let } \lim_{x \rightarrow 2^-} f(x) = f(2), \lim_{x \rightarrow 3^-} f(x) = f(3) \rightarrow a = \frac{1}{3}, b = \frac{1}{6}$$

Find a and b that make $f(x) = \lim_{n \rightarrow +\infty} \frac{x^{2n-1} + ax^2 + bx}{x^{2n} + 1}$ continuous on $(-\infty, \infty)$.

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Solution:

$$f(x) = \begin{cases} ax^2 + bx : -1 < x < 1 \\ \frac{a - b - 1}{2} : x = -1 \\ \frac{a + b + 1}{2} : x = 1 \\ \frac{1}{x} : x > 1 \text{ or } x < -1 \end{cases}$$

$$\text{At } x = 1, \begin{cases} \frac{a + b + 1}{2} = a + b \\ a + b = 1 \end{cases}$$

$$\text{At } x = -1, a - b = -1$$

Thus, we have $a = 0, b = 1$

There are two functions:

$f(x)$ is continuous on $(-\infty, \infty)$, and $f(x) \neq 0$.

$\varphi(x)$ is defined on $(-\infty, \infty)$, but $\varphi(x)$ has discontinuity.

Judge whether the following four statements are correct:

- ① $\varphi[f(x)]$ must have discontinuity.
- ② $[\varphi(x)]^2$ must have discontinuity.
- ③ Whether $f[\varphi(x)]$ has discontinuity is uncertain.
- ④ $\frac{\varphi(x)}{f(x)}$ must have discontinuity.

① **No.** $\varphi[f(x)]$ is not necessarily discontinuous.

- **Counterexample:** Let $f(x) = c$ (a constant). The range of f is a single point. If $\varphi(x)$ happens to be continuous at that specific point c , then the composite function $\varphi[f(x)] = \varphi(c)$ is a constant, and therefore continuous everywhere.

② **No.** $[\varphi(x)]^2$ is not necessarily discontinuous.

- **Counterexample:** Consider the sign function, which is discontinuous at $x = 0$:

$$\varphi(x) = \begin{cases} 1 & x \geq 0 \\ -1 & x < 0 \end{cases}$$

However, its square, $[\varphi(x)]^2 = 1$, is a constant function and is continuous everywhere.

③ **Yes.** The continuity of $f[\varphi(x)]$ is uncertain.

- It depends on the functions. If $f(x)$ is a constant function, $f(x) = c$, then $f[\varphi(x)] = c$ is continuous.
- However, if $f(x) = x$ (which is continuous), then $f[\varphi(x)] = \varphi(x)$, which is discontinuous by definition.
- Since we can find cases where it's continuous and cases where it's not, the outcome is uncertain.

④ **Yes.** $\frac{\varphi(x)}{f(x)}$ must have a discontinuity.

- **Proof by Contradiction:** Let $h(x) = \frac{\varphi(x)}{f(x)}$. Assume that $h(x)$ is continuous.
- We can rearrange the equation to $\varphi(x) = h(x) \cdot f(x)$.
- This would mean $\varphi(x)$ is the product of two continuous functions, which must also be continuous.
- This contradicts our premise that $\varphi(x)$ has a discontinuity. Therefore, our assumption must be false, and $h(x)$ must be discontinuous.

There are two functions:

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Solution:

- ① No
- ② No
- ③ Yes
- ④ Yes

$$\textcircled{1} \lim_{x \rightarrow a} c = c$$

$$\textcircled{2} \lim_{x \rightarrow a} x^n = a^n$$

$$\textcircled{3} \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$\textcircled{4} \lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$$

$$\textcircled{5} \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$$

$$\textcircled{6} \lim_{x \rightarrow 0} \frac{\arcsin x}{x} = 1$$

$$\textcircled{7} \lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$$

$$\textcircled{8} \lim_{x \rightarrow 0^+} x^x = 1$$

$$\textcircled{1} \lim_{x \rightarrow 0^+} x \ln x = 0$$

$$\textcircled{2} \lim_{x \rightarrow 0} \frac{\sqrt[n]{x+1} - 1}{x} = \frac{1}{n}$$

$$\textcircled{3} \lim_{x \rightarrow \infty} x^{\frac{1}{x}} = 1$$

$$\textcircled{4} \lim_{x \rightarrow \pm\infty} \left(1 + \frac{1}{x}\right)^x = e$$

$$\textcircled{5} \lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} = e$$

$$\textcircled{6} \lim_{x \rightarrow 0} (1+\sin x)^{\frac{1}{x}} = e$$

$$\textcircled{7} \lim_{x \rightarrow \infty} \left(x - \sqrt{x^2 - a^2}\right) = 0$$

$$\textcircled{8} \lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1$$

2. Limits

Suppose $f(x)$ is defined when x is near the number a . (This means that f is defined on some open interval that contains a , **except possibly** at a itself.)

Then we write

$$\lim_{x \rightarrow a} f(x) = L$$

and say

"the limit of $f(x)$, as x approaches a , equals L "

if we can make the values of $f(x)$ arbitrarily close to L (as close to L as we like) by taking x to be **sufficiently close to** a (on either side of a) but **not equal to** a .

Considering a function called *Heaviside Function*

$$H(t) = \begin{cases} 0 & t < 0 \\ 1 & t \geq 0 \end{cases}$$

Does $\lim_{t \rightarrow 0} H(t)$ exists?

We write

$$\lim_{x \rightarrow a^-} f(x) = L$$

and say the left-hand limit of $f(x)$ as x approaches a is equal to L if we can make the values of $f(x)$ arbitrarily close to L by taking x to be **sufficiently close to a** and x less than a .

When calculating $\lim_{x \rightarrow a^-} f(x)$, we consider only $x < a$.

Similarly, we can get the right-hand limit of $f(x)$ as x approaches a .

When does $\lim_{x \rightarrow a} f(x)$ exist?

$$\lim_{x \rightarrow a} f(x) = L$$

if and only if

$$\lim_{x \rightarrow a^-} f(x) = L \text{ and } \lim_{x \rightarrow a^+} f(x) = L$$

Can we directly regard L as $f(a)$?

Let f be a function defined on both sides of a , **except possibly** at a itself. Then

$$\lim_{x \rightarrow a} f(x) = \infty$$

means that the values of $f(x)$ can be made arbitrarily large (as large as we please) by taking x **sufficiently close to** a , but **not equal to** a .

Let f be a function defined on both sides of a , **except possibly** at a itself. Then

$$\lim_{x \rightarrow a} f(x) = -\infty$$

means that the values of $f(x)$ can be made arbitrarily large negative by taking x **sufficiently close to** a , but **not equal to** a .

Warning:

$\lim_{x \rightarrow a} f(x) = (-)\infty$ does not mean that we are regarding $(-)\infty$ as a number. Nor does it mean that the limit exists!

- ① Let f be a function defined on some interval (a, ∞) . Then

$$\lim_{x \rightarrow \infty} f(x) = L$$

means that the values of $f(x)$ can be made arbitrarily close to L by taking x sufficiently large.

- ② Let f be a function defined on some interval $(-\infty, a)$. Then

$$\lim_{x \rightarrow -\infty} f(x) = L$$

means that the values of $f(x)$ can be made arbitrarily close to L by taking x sufficiently large negative.

Note: Limits that involve ∞ 's are always **one-sided**.

$$\textcircled{1} \quad \lim_{x \rightarrow \infty} f(x) = \infty$$

$$\textcircled{2} \quad \lim_{x \rightarrow \infty} f(x) = -\infty$$

$$\textcircled{3} \quad \lim_{x \rightarrow -\infty} f(x) = \infty$$

$$\textcircled{4} \quad \lim_{x \rightarrow -\infty} f(x) = -\infty$$

Derivative as limit

The tangent line to the curve $y = f(x)$ at the point $(a, f(a))$ is the line through P with slope

$$m = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h}$$

Tangent line as limit

The curve has a vertical tangent line when $x = a$; that is, f is continuous at a and

$$\lim_{x \rightarrow a} |f'(x)| = \infty$$

Five basic laws

Suppose that c is a constant and the limits

$$\lim_{x \rightarrow a} f(x) \text{ and } \lim_{x \rightarrow a} g(x)$$

exists.

Then

$$\textcircled{1} \lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$$

$$\textcircled{2} \lim_{x \rightarrow a} [f(x) - g(x)] = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x)$$

$$\textcircled{3} \lim_{x \rightarrow a} [cf(x)] = c \lim_{x \rightarrow a} f(x)$$

$$\textcircled{4} \lim_{x \rightarrow a} [f(x)g(x)] = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x)$$

$$\textcircled{5} \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} \quad (\text{if } \lim_{x \rightarrow a} g(x) \neq 0)$$

Another six laws:

$$\textcircled{1} \lim_{x \rightarrow a} [f(x)]^n = [\lim_{x \rightarrow a} f(x)]^n \text{ (where } n \text{ is a positive integer)}$$

$$\textcircled{2} \lim_{x \rightarrow a} c = c$$

$$\textcircled{3} \lim_{x \rightarrow a} x = a$$

$$\textcircled{4} \lim_{x \rightarrow a} x^n = a^n \text{ (where } n \text{ is a positive integer)}$$

$$\textcircled{5} \lim_{x \rightarrow a} \sqrt[n]{x} = \sqrt[n]{a} \text{ (where } n \text{ is a positive integer)}$$

if n is even, we assume that $a > 0$

$$\textcircled{6} \lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \rightarrow a} f(x)} \text{ (where } n \text{ is a positive integer)}$$

if n is even, we assume that $\lim_{x \rightarrow a} f(x) > 0$

For composite functions, we have

Law for Composite Functions

$$\lim_{x \rightarrow a} f(g(x)) = f(\lim_{x \rightarrow a} g(x))$$

for all the limits and functions exist.

E.g. Solve $\lim_{x \rightarrow \infty} \cos \frac{1}{x}$.

Warning:

The Limit Laws can't be applied to infinite limits because $(-)\infty$ is not a number!

That is, when performing operations on limits, it must be ensured that the limits involved in the operation all exist (i.e., are not indeterminate forms or infinity) and are finite values.

Two additional properties of limits:

- ① if $f(x) \leq g(x)$ when x is near a (**except possibly** at a) and the limits of f and g both exist as x approaches a , then

$$\lim_{x \rightarrow a} f(x) \leq \lim_{x \rightarrow a} g(x)$$

- ② (**The Squeeze Theorem**) if $f(x) \leq g(x) \leq h(x)$ when x is near a (**except possibly** at a) and

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L$$

then

$$\lim_{x \rightarrow a} g(x) = L$$

Common Methods for Calculating Limits:

- ① Basic Arithmetic and Function Properties
- ② Definition of the Derivative
- ③ **Two Important Limits**
- ④ **L'Hôpital's Rule**
- ⑤ *Equivalent Infinitesimals*
- ⑥ *Taylor Expansion (For reference only)*

Basic Arithmetic and Function Properties

Limit of Rational Functions:

For the rational function $F(x) = \frac{p(x)}{q(x)} = \frac{a_0 + a_1x_1 + \cdots + a_mx^m}{b_0 + b_1x_1 + \cdots + b_nx^n}$, we have

$$\lim_{x \rightarrow \infty} F(x) = \begin{cases} \frac{a_m}{b_n} & m = n \\ 0 & m < n \\ \infty & m > n \end{cases}$$

1. Calculate the following limits:

$$\textcircled{1} \lim_{x \rightarrow 0} \frac{4x^3 - 2x^2 + x}{2x + 3x^2}$$

$$\textcircled{2} \lim_{x \rightarrow \infty} \frac{3x^3 + 4x^2 + 2}{7x^3 + 5x^2 - 3}$$

$$\textcircled{3} \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right) \left(2 - \frac{1}{x^2}\right)$$

2. Let $\lim_{x \rightarrow -1} \frac{x^3 - ax^2 - x + 4}{x + 1}$ have a limit value m . Find the values of a and m .

$$\textcircled{1} \lim_{x \rightarrow 0} \frac{4x^3 - 2x^2 + x}{2x + 3x^2} = \lim_{x \rightarrow 0} \frac{4x^2 - 2x + 1}{2 + 3x} = 0.5$$

$$\textcircled{2} \lim_{x \rightarrow \infty} \frac{3x^3 + 4x^2 + 2}{7x^3 + 5x^2 - 3} = \lim_{x \rightarrow \infty} \frac{3 + \frac{4}{x} + \frac{2}{x^3}}{7 + \frac{5}{x} - \frac{3}{x^3}} = \frac{3}{7}$$

$$\textcircled{3} \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right) \left(2 - \frac{1}{x^2}\right) = \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right) \cdot \lim_{x \rightarrow \infty} \left(2 - \frac{1}{x^2}\right) = 1 \cdot 2 = 2$$

$\textcircled{4}$ Since the denominator $\rightarrow 0$ as $x \rightarrow -1$, the numerator must also be 0 for the limit to exist.

$$\implies \lim_{x \rightarrow -1} (x^3 - ax^2 - x + 4) = (-1)^3 - a(-1)^2 - (-1) + 4 = 4 - a = 0 \implies a = 4.$$

Now, we find the limit m :

$$m = \lim_{x \rightarrow -1} \frac{x^3 - 4x^2 - x + 4}{x + 1} = \lim_{x \rightarrow -1} \frac{(x + 1)(x^2 - 5x + 4)}{x + 1} = \lim_{x \rightarrow -1} (x^2 - 5x + 4) = (-1)^2 - 5(-1) + 4 = 10.$$

Thus, $a = 4$ and $m = 10$.

Involving Roots: Rationalize the Numerator or Substitute the Root Expression

3. Calculate the following limits:

$$\textcircled{1} \lim_{x \rightarrow 0} \frac{-1 + \sqrt[n]{x+1}}{x}$$

$$\textcircled{2} \lim_{x \rightarrow -4} \frac{\sqrt{9+x^2} - 5}{x+4}$$

$$\textcircled{3} \lim_{x \rightarrow -\infty} (\sqrt{x^2 + 2x} + x)$$

$$\textcircled{4} \lim_{x \rightarrow 0} \frac{\sqrt{1 + \tan^2 x} - \sqrt{1 + \sin^2 x}}{(5^x - 1) \arctan^3 x}$$

$$\textcircled{1} \text{ Let } t^n - 1 := x, \lim_{x \rightarrow 0} \frac{-1 + \sqrt[n]{x+1}}{x} = \lim_{t \rightarrow 1} \frac{t-1}{t^n-1} = \lim_{t \rightarrow 1} \frac{1}{1+t+t^2+\dots+t^{n-1}} = \frac{1}{n}$$

$$\textcircled{2} \lim_{x \rightarrow -4} \frac{\sqrt{9+x^2}-5}{x+4} = \lim_{x \rightarrow -4} \frac{x^2-16}{(x+4)(\sqrt{9+x^2}+5)} = \lim_{x \rightarrow -4} \frac{x-4}{\sqrt{9+x^2}+5} = -\frac{4}{5}$$

$$\textcircled{3} \lim_{x \rightarrow -\infty} (\sqrt{x^2+2x}+x) = \lim_{x \rightarrow -\infty} \frac{2x}{\sqrt{x^2+2x}-x} = \lim_{x \rightarrow -\infty} \frac{2x}{-x\sqrt{1+2/x}-x} =$$

$$\lim_{x \rightarrow -\infty} \frac{2}{-\sqrt{1+2/x}-1} = -1$$

$$\textcircled{4} \lim_{x \rightarrow 0} \frac{\sqrt{1+\tan^2 x} - \sqrt{1+\sin^2 x}}{(5^x-1)\arctan^3 x} = \lim_{x \rightarrow 0} \frac{\frac{1}{2}(\tan^2 x - \sin^2 x)}{(x \ln 5)(x^3)} = \lim_{x \rightarrow 0} \frac{\frac{1}{2}x^4}{x^4 \ln 5} = \frac{1}{2 \ln 5}$$

$$\textcircled{1} \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

Relevant transformation:

- $\lim_{x \rightarrow \infty} \frac{\sin x}{x} = 0$
- $\lim_{f(x) \rightarrow 0} \frac{\sin f(x)}{f(x)} = 1$

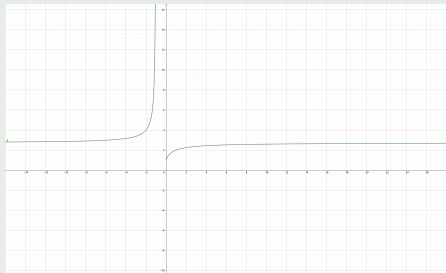
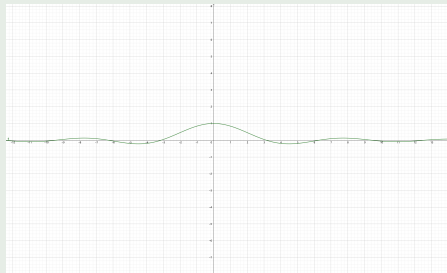
Be sure to keep these two limits in mind!

① $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$

(How to prove it? Considering $\sin x$, x and $\tan x$. Then, use the squeeze theorem.)

② $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e = 2.718281828459045 \dots$

(Proof: <https://www.zhihu.com/question/277272238>)



$$\textcircled{1} \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e$$

Relevant transformation:

- $\lim_{x \rightarrow 0^+} \left(1 + \frac{1}{x}\right)^x = 1$
- $\lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} = e$
- $\lim_{x \rightarrow +\infty} (1+x)^{\frac{1}{x}} = 1$
- $\lim_{x \rightarrow 0^-} \left(1 - \frac{1}{x}\right)^x = 1$
- $\lim_{x \rightarrow \infty} \left(1 - \frac{1}{x}\right)^x = \frac{1}{e}$

If $u \rightarrow 1$, $v \rightarrow \infty$, then we have

$$\lim u^v = "1^\infty" = e^{\lim v(u-1)}$$

L'Hospital's Rule

Suppose f and g are differentiable and $g'(x) \neq 0$ on an open interval I that contains a (except possibly at a). Suppose that

$$\lim_{x \rightarrow a} f(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = 0$$

or that

$$\lim_{x \rightarrow a} f(x) = \pm\infty \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = \pm\infty$$

(In other words, we have an indeterminate form of type $\frac{0}{0}$ or ∞/∞ .) Then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

if the limit on the right side exists (or is ∞ or $-\infty$).

The Challenge

When we have a limit of the form $\lim f(x)g(x)$ where $f(x) \rightarrow 0$ and $g(x) \rightarrow \infty$, we cannot simply multiply the limits. We must first convert the expression into a fraction to use L'Hôpital's Rule.

Strategy: Invert and Convert

Rewrite the product $f(x)g(x)$ into one of two fractional forms:

$$f(x)g(x) = \frac{f(x)}{1/g(x)} \quad \left(\text{yields } \frac{0}{0} \right) \quad \text{or} \quad f(x)g(x) = \frac{g(x)}{1/f(x)} \quad \left(\text{yields } \frac{\infty}{\infty} \right)$$

Tip: Choose the conversion that results in an easier derivative. Usually, converting the simpler function (e.g., a polynomial) is best.

The Challenge

A limit of the form $\lim(f(x) - g(x))$ where both $f(x) \rightarrow \infty$ and $g(x) \rightarrow \infty$ is indeterminate. The result depends entirely on the relative rates at which the functions approach infinity.

Strategy: Combine into a Single Fraction

The goal is to perform an algebraic manipulation that turns the difference into a single quotient. This allows you to then check if L'Hôpital's Rule applies. Common methods include:

- Finding a common denominator.
- Rationalizing by multiplying by a conjugate.

The Challenge

Limits of the form $\lim f(x)^{g(x)}$ can be indeterminate. For example, 1^∞ is not necessarily 1 because the base is only *approaching* 1 while the exponent is growing.

Strategy: The Logarithmic Method

Since L'Hôpital's rule doesn't work on exponents directly, we use logarithms to bring the exponent down and convert the problem into a product.

- ➊ **Set up:** Let $y = \lim f(x)^{g(x)}$.
- ➋ **Take Logarithm:** Consider $\ln(y) = \lim \ln(f(x)^{g(x)}) = \lim g(x) \ln(f(x))$.
- ➌ **Solve New Limit:** This new limit will be a $0 \cdot \infty$ form. Solve it for a value L .
- ➍ **Exponentiate:** The original answer is $y = e^L$.

Calculate the following limits, which may require algebraic manipulation before applying L'Hôpital's Rule.

① $\lim_{x \rightarrow 0^+} \left(\frac{1}{x} \right)^{\tan x}$ (∞^0 type with a twist)

② $\lim_{x \rightarrow 0} \left(\frac{1}{\ln(1+x)} - \frac{1}{x} \right)$ ($\infty - \infty$ type requiring combination)

③ $\lim_{x \rightarrow \infty} x \left(\arctan(x) - \frac{\pi}{2} \right)$ ($\infty \cdot 0$ type needing careful rewriting)

① Let $y = (\frac{1}{x})^{\tan x}$, so $\ln y = -\tan x \ln x = -\frac{\ln x}{\cot x}$.

$$\lim_{x \rightarrow 0^+} \ln y \stackrel{L'H}{=} \lim_{x \rightarrow 0^+} \frac{1/x}{-\csc^2 x} = \lim_{x \rightarrow 0^+} \frac{\sin^2 x}{x} = 0.$$

The limit is $e^0 = 1$.

$$\begin{aligned} \textcircled{2} \quad L &= \lim_{x \rightarrow 0} \frac{x - \ln(1+x)}{x \ln(1+x)} \stackrel{L'H}{=} \lim_{x \rightarrow 0} \frac{1 - 1/(1+x)}{\ln(1+x) + x/(1+x)} = \lim_{x \rightarrow 0} \frac{x}{(1+x) \ln(1+x) + x} \\ &\stackrel{L'H}{=} \lim_{x \rightarrow 0} \frac{1}{\ln(1+x) + 1 + 1} = \frac{1}{2}. \end{aligned}$$

$$\textcircled{3} \quad \text{Let } t = 1/x. \text{ The limit is } \lim_{t \rightarrow 0^+} \frac{\arctan(1/t) - \pi/2}{t} = \lim_{t \rightarrow 0^+} \frac{-\arctan(t)}{t}.$$

$$\stackrel{L'H}{=} \lim_{t \rightarrow 0^+} \frac{-1/(1+t^2)}{1} = -1.$$

3. Equivalent Infinitesimal and Taylor Expansion

When $x \rightarrow 0$

$$\tan x - \sin x \sim \frac{x^3}{2}$$

$$\tan x - x \sim \frac{x^3}{3}$$

$$x - \arctan x \sim \frac{x^3}{3}$$

$$x - \sin x \sim \frac{x^3}{6}$$

$$\arcsin x - x \sim \frac{x^3}{6}$$

$$\textcircled{1} \lim_{x \rightarrow 0} \frac{\sin \omega x}{x}$$

$$\textcircled{2} \lim_{x \rightarrow \infty} \left(1 - \frac{1}{x}\right)^{kx} \quad (k \text{ is a positive integer})$$

$$\textcircled{3} \lim_{x \rightarrow \frac{\pi}{2}} (\sin x)^{\tan x}$$

$$\textcircled{4} \lim_{x \rightarrow 0^+} (\cos \sqrt{x})^{\frac{\pi}{x}}$$

$$\textcircled{5} \lim_{x \rightarrow \infty} \left(\frac{x+3}{x+6}\right)^{\frac{x-1}{2}}$$

$$\textcircled{1} \lim_{x \rightarrow 0} \frac{\sin \omega x}{x} = \lim_{x \rightarrow 0} \frac{\omega x}{x} = \omega$$

$$\textcircled{2} \lim_{x \rightarrow \infty} \left(1 - \frac{1}{x}\right)^{kx} = \lim_{x \rightarrow \infty} \left[\left(1 + \frac{1}{-x}\right)^{-x}\right]^{-k} = e^{-k}$$

$$\textcircled{3} \text{ Let } y = x - \frac{\pi}{2}. \text{ The limit is } \lim_{y \rightarrow 0} (\cos y)^{-1/\tan y}.$$

$$\text{Using } e^{\lim v(u-1)} \implies \exp \left[\lim_{y \rightarrow 0} \frac{-1}{\tan y} (\cos y - 1) \right] = \exp \left[\lim_{y \rightarrow 0} \frac{-1}{y} \left(-\frac{y^2}{2} \right) \right] = e^0 = 1$$

$$\textcircled{4} \text{ Using } e^{\lim v(u-1)} \implies \exp \left[\lim_{x \rightarrow 0^+} \frac{\pi}{x} (\cos \sqrt{x} - 1) \right] =$$

$$\exp \left[\lim_{x \rightarrow 0^+} \frac{\pi}{x} \left(-\frac{(\sqrt{x})^2}{2} \right) \right] = e^{-\pi/2}$$

5 Using $e^{\lim v(u-1)} \implies \exp \left[\lim_{x \rightarrow \infty} \frac{x-1}{2} \left(\frac{x+3}{x+6} - 1 \right) \right] =$

$$\exp \left[\lim_{x \rightarrow \infty} \frac{x-1}{2} \left(\frac{-3}{x+6} \right) \right] = e^{-3/2}$$

Example: Solve the limit

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{\ln(1 + 4x)}{\sin(3x)} &= \lim_{x \rightarrow 0} \frac{\ln(1 + 4x)}{\sin(3x)} \lim_{x \rightarrow 0} \frac{4x}{\ln(1 + 4x)} \lim_{x \rightarrow 0} \frac{\sin(3x)}{3x} \\ &= \lim_{x \rightarrow 0} \frac{4x}{3x} = 4/3\end{aligned}$$

Definition

Taylor expansion around $x = x_0$:

$$f(x) = f(x_0) + \sum_{i=1}^n \frac{f^{(i)}(x_0)}{i!} (x - x_0)^i + R_n, \text{ where } R_n = o[(x - x_0)^n]$$

It simulates a function around a point with a polynomial function.

Taylor expansion of some polynomials when x is around 0:

$$\textcircled{1} e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + o(x^3)$$

$$\textcircled{2} \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} + o(x^3)$$

$$\textcircled{3} \sin x = x - \frac{x^3}{6} + \frac{x^5}{120} + o(x^5)$$

$$\textcircled{4} \cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} + o(x^4)$$

$$\textcircled{5} \tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + o(x^5)$$

Tip

$o(x^n)$ means the order of the polynomial is **larger** than n ;

$O(x^n)$ means the order of the polynomial is **larger than or equal to** n .

The transformation of Taylor Expansion:

Example

The Taylor expansion of e^{x^2} around $x=0$:

$$e^{x^2} = 1 + x^2 + \frac{(x^2)^2}{2} + \frac{(x^2)^3}{6} + o((x^2)^3) = 1 + x^2 + \frac{x^4}{2} + \frac{x^6}{6} + o(x^6)$$

Some methods to calculate the limits:

- ① Use those limit laws directly
- ② Exchange the order of functions and limit symbols based on the continuity of composite function. (Will be mentioned later)
- ③ Do factorization, denominator rationalization or numerator rationalization.
- ④ If a factor approaching zero is found in the denominator, try to eliminate it.
- ⑤ Translate the formula into the form of "two important limits"

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